Year 8
Mathematics

Unit 4: 2D Shapes and 3D Solids

4.1 Area of Triangles, Parallelograms and Trapezia

Area of Triangles, Parallelograms and Trapezia

STARTER · 5 MIN

Date: _____ **Title: Area of Triangles, Parallelograms and Trapezia**

LEARNING OBJECTIVES

- Use the formula for the area of a triangle, parallelogram and trapezium

STARTER *Answer in your book. No calculators.*

1. Area of a rectangle 7 cm by 4 cm

2. Work out $\frac{1}{2} \times 10 \times 6$

3. Work out $(9 + 15) \div 2$

4. Work out $54 \div 6$

Starter — Answers

STARTER

Tick or correct in green pen

1. **28 cm²**
2. **30**
3. **12**
4. **9**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can substitute values into area formulas and give units	☐
SILVER	I can solve real-life problems involving area	☐
GOLD	I can rearrange a formula to find a missing measurement	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Area of Triangles, Parallelograms and Trapezia

The height is always perpendicular (at 90°) to the base.

Parallelogram

Area = base × height

$A = bh$

Triangle

Half of a parallelogram.

$$A = \frac{1}{2}bh$$

Trapezium

Add the parallel sides, halve, × height.

$$A = \frac{1}{2}(a + b)h$$

Units

Area is in square units.

cm^2, m^2, mm^2

Examples 1 and 2 — Parallelograms

EXAMPLES · 5 MIN

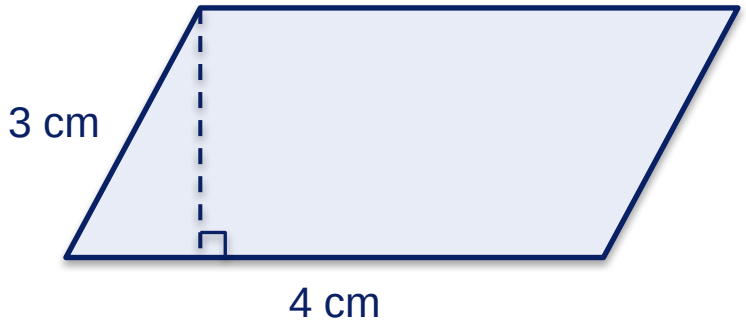
BRONZE

Watch and copy into your book

Hint
Parallelogram: base × height
Triangle: base × height ÷ 2

Example 1

Find the area.



Example 2

The area is 54 cm^2.

Find the height h.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Area = base × height

$A = 4 \times 3$

$= \mathbf{12\ cm^2}$

Tip: use the height, not the sloping side.

Example 2

Base × height = area

$6 \times h = 54$

$h = 54 \div 6$

$= \mathbf{9\ cm}$

Tip: divide the area by the base.

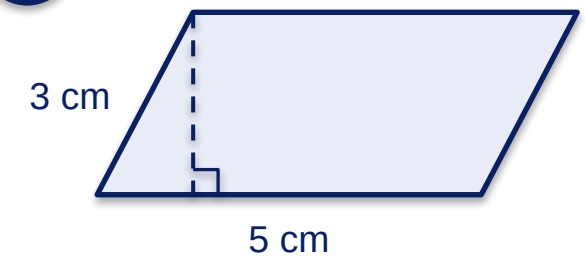
Task 1

PRACTICE · 9 MIN

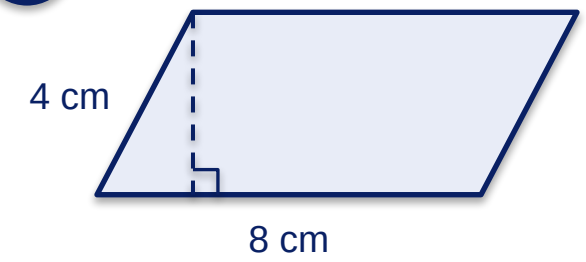
BRONZE *Find the area*

Hint Parallelogram: base × height Triangle: base × height ÷ 2

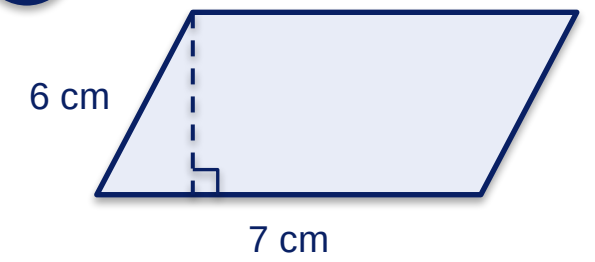
1 Area



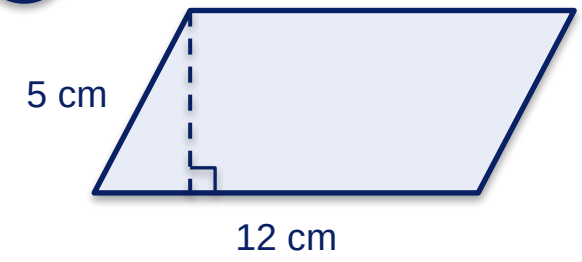
2 Area



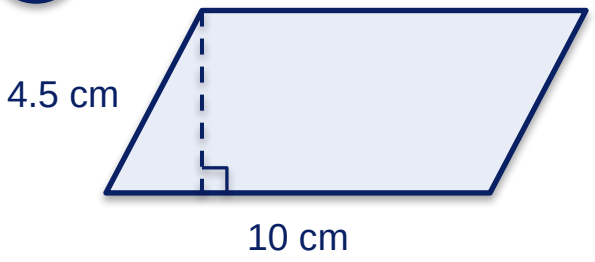
3 Area



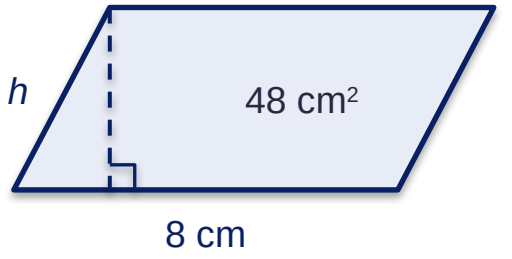
4 Area



5 Area



★ Find *h*



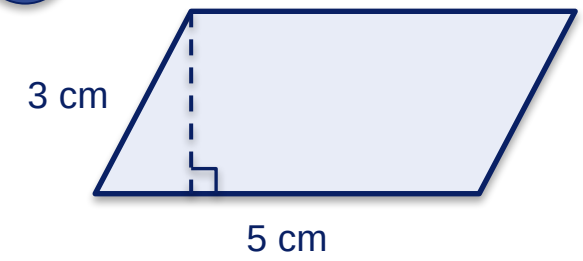
Task 1 — Answers

CORRECTIONS

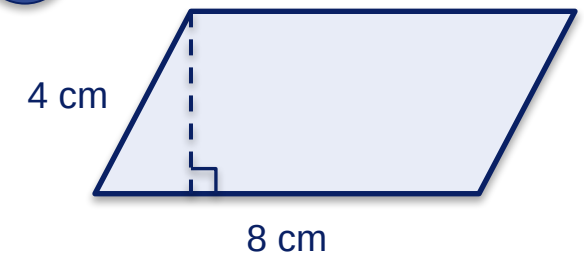
BRONZE

Tick or correct in green pen

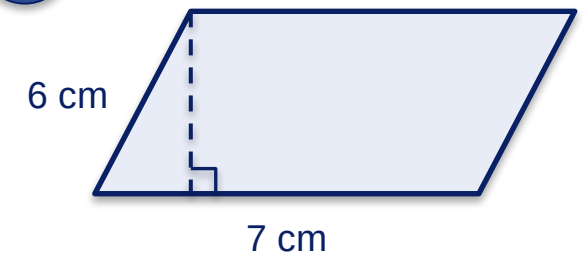
1 **15 cm²**



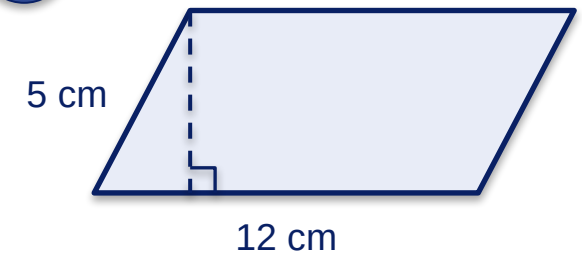
2 **32 cm²**



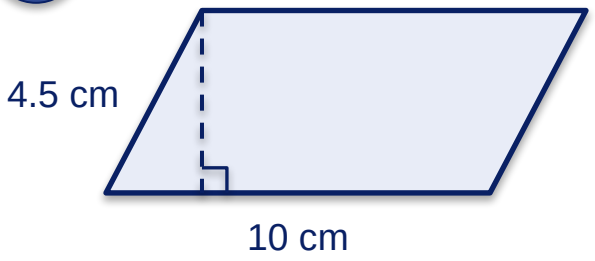
3 **42 cm²**



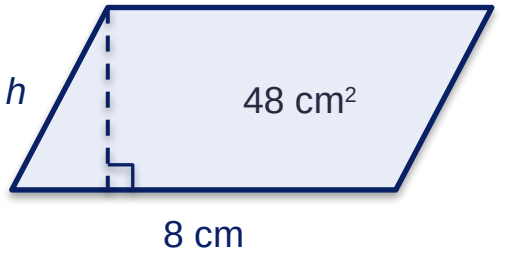
4 **60 cm²**



5 **45 cm²**



★ ***h* = 6 cm**



Examples 3 and 4 — Triangles

EXAMPLES · 5 MIN

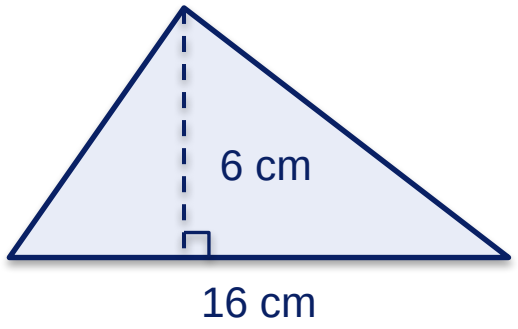
SILVER

Watch and copy into your book

Hint Use the perpendicular height Trapezium: $(a + b) \div 2 \times h$

Example 3

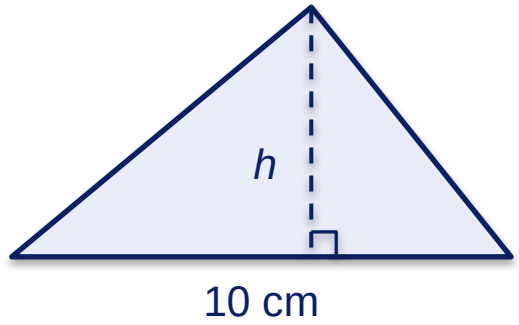
Find the area.



Example 4

The area is 35 cm^2.

Find the height h.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Half of base × height

$$A = \frac{1}{2} \times 16 \times 6$$

$$= 8 \times 6$$

$$\mathbf{= 48\ cm^2}$$

Tip: halve the base first: 16 ÷ 2 = 8.

Example 4

Write an equation

$$\frac{1}{2} \times 10 \times h = 35$$

$$5h = 35$$

$$\mathbf{h = 7\ cm}$$

Tip: half of 10 is 5.

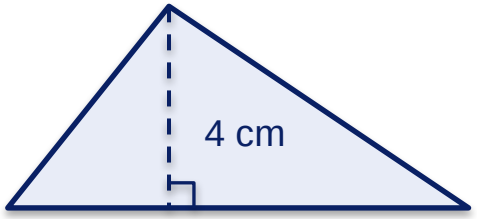
Task 2

PRACTICE · 9 MIN

SILVER *Find the area*

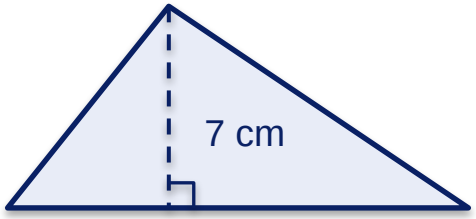
Hint Use the perpendicular height Trapezium: $(a + b) \div 2 \times h$

1 Area



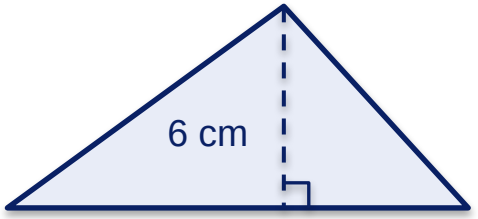
10 cm

2 Area



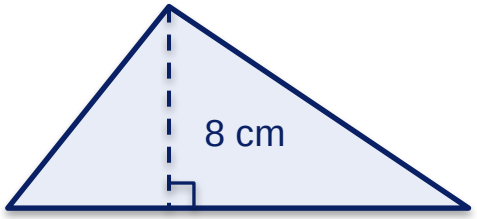
12 cm

3 Area



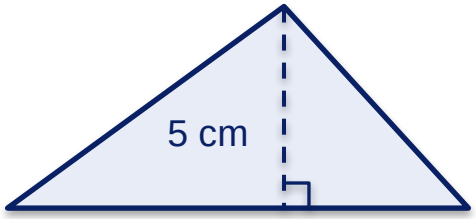
9 cm

4 Area



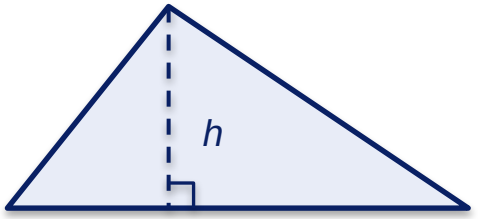
15 cm

5 Area



7 cm

★ Area 40 cm². Find *h*



10 cm

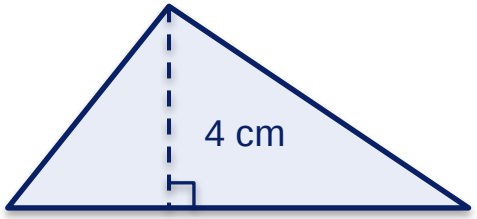
Task 2 — Answers

CORRECTIONS

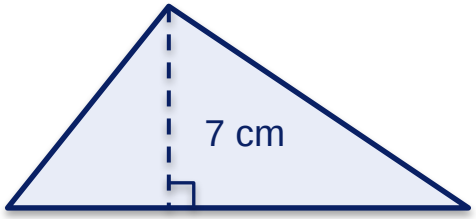
SILVER

Tick or correct in green pen

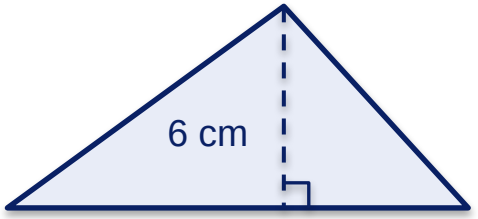
1 **20 cm²**



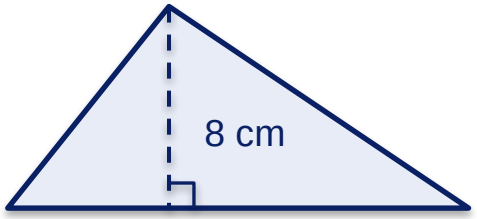
2 **42 cm²**



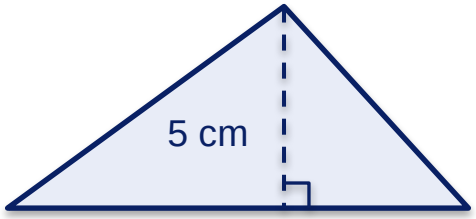
3 **27 cm²**



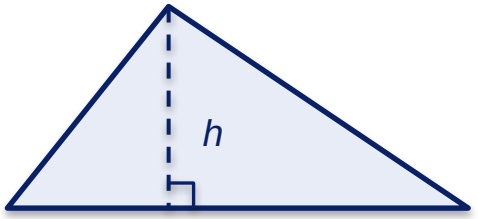
4 **60 cm²**



5 **17.5 cm²**



★ ***h* = 8 cm**



Examples 5 and 6 — Trapezia

EXAMPLES · 5 MIN

GOLD

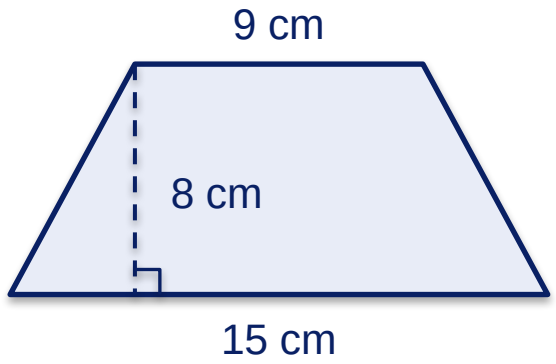
Exam-style: show all your working

Hint Area first, then cost Trapezium: $(a + b) \div 2 \times h$

Example 5

(2 marks)

Find the area.



Example 6

(3 marks)

A windscreen is a trapezium.

Parallel sides: 1.0 m and 1.2 m

Height: 0.7 m

Glass costs £325 per m^2.

Find the cost of the glass.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Add the parallel sides first

$$A = \frac{1}{2} \times (9 + 15) \times 8$$

$$= \frac{1}{2} \times 24 \times 8$$

$$\mathbf{= 96\ cm^2}$$

Tip: 9 + 15 = 24, half of 24 is 12.

Example 6

Area, then cost

$$A = \frac{1}{2} \times (1.0 + 1.2) \times 0.7$$

$$= 0.77\ m^2$$

$$\text{cost} = 0.77 \times 325$$

$$\mathbf{= £250.25}$$

Tip: money has 2 decimal places.

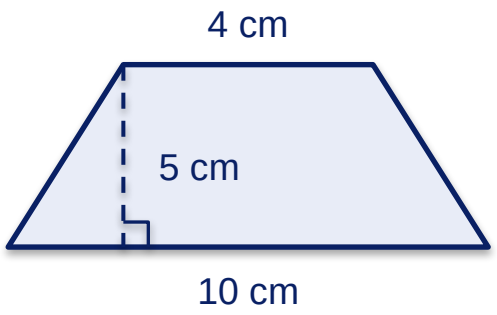
Task 3

PRACTICE · 9 MIN

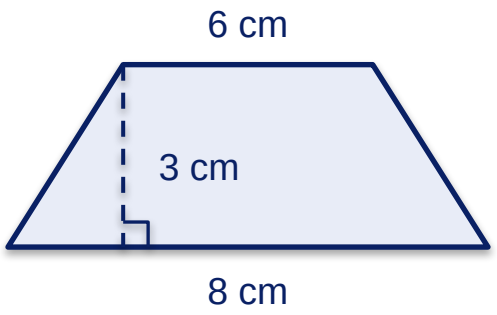
GOLD *Find the area*

Hint Area first, then cost Trapezium: $(a + b) \div 2 \times h$

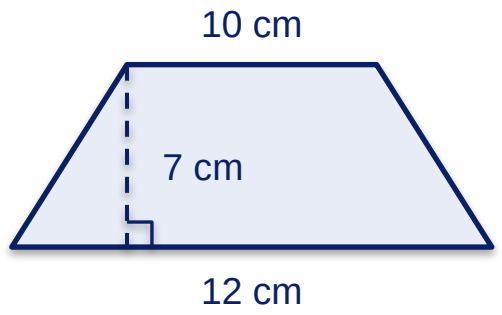
1 Area



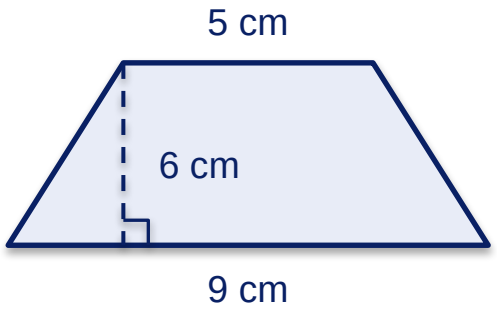
2 Area



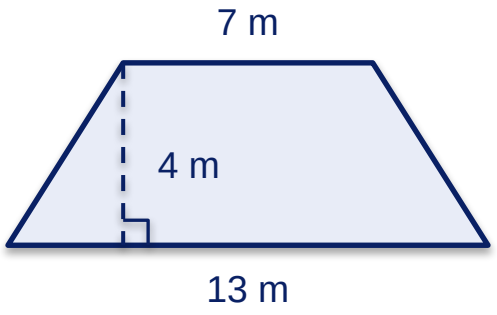
3 Area



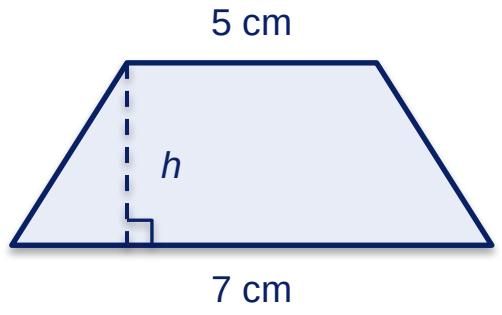
4 Area



5 Area



★ Area 60 cm². Find *h*



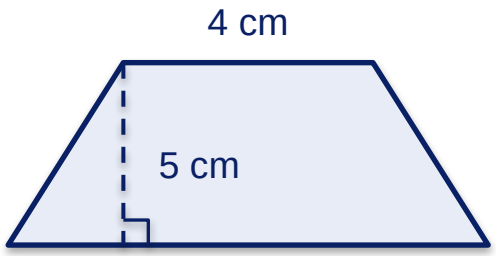
Task 3 — Answers

CORRECTIONS

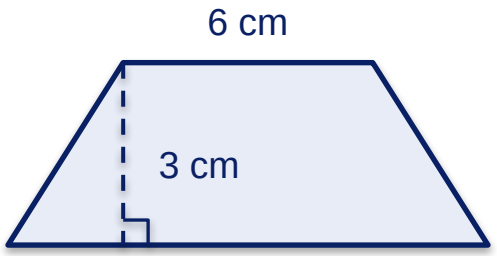
GOLD

Tick or correct in green pen

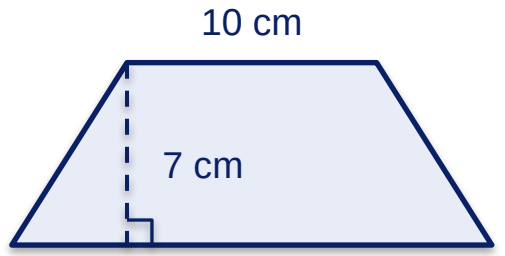
1 **35 cm²**



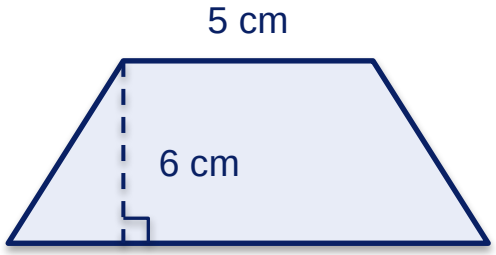
2 **21 cm²**



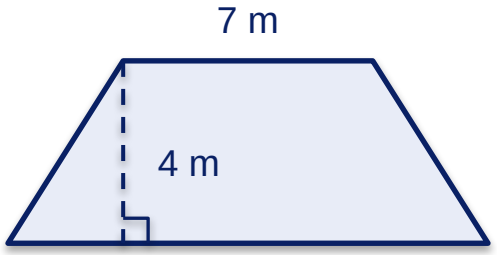
3 **77 cm²**



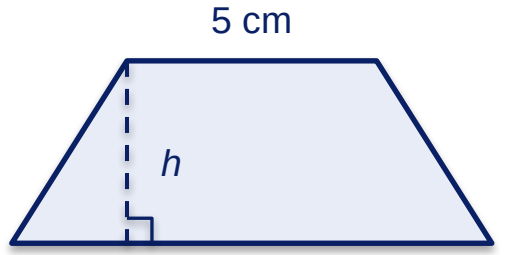
4 **42 cm²**



5 **40 m²**



★ ***h* = 10 cm**



Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Formula for the area of a parallelogram

SILVER Triangle: base 6 cm, height 9 cm. Area?

GOLD Which two lengths do you add for a trapezium?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **A = base × height**

SILVER **27 cm²**

GOLD **The two parallel sides**

Exam Practice — Parallelograms and Triangles

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

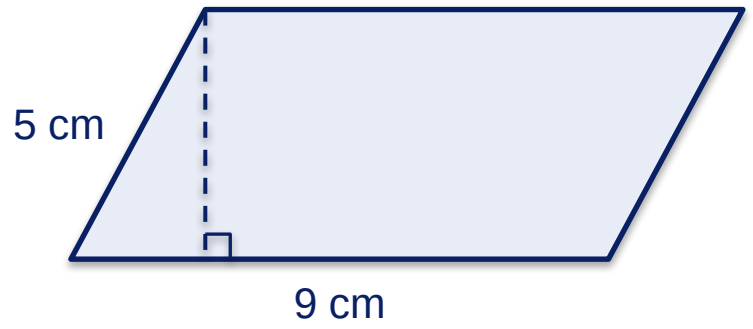
Hint
Parallelogram: base × height
Triangle: base × height ÷ 2

Question 1

(2 marks)

Work out the area of this parallelogram.

Diagram NOT accurately drawn



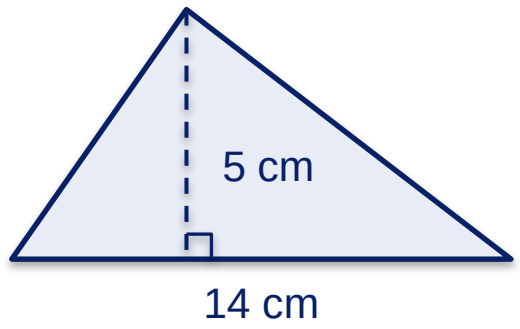
..... cm^2

Question 2

(2 marks)

Work out the area of this triangle.

Diagram NOT accurately drawn



..... cm^2

Exam Practice — Trapezium and Missing Height

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

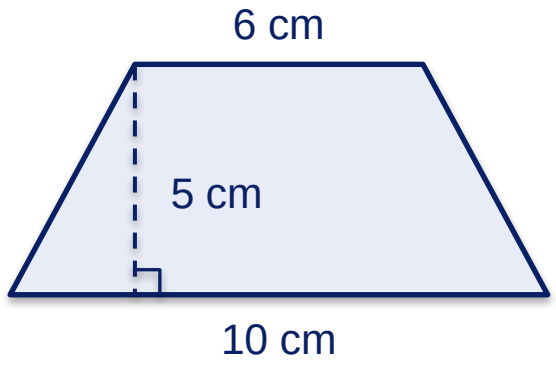
Hint Use the perpendicular height
Trapezium: $(a + b) \div 2 \times h$

Question 3

(2 marks)

Work out the area of this trapezium.

Diagram NOT accurately drawn



..... cm^2

Question 4

(2 marks)

A triangle has an area of 36 cm^2.
Its base is 8 cm.

Work out its height.

..... cm

Exam Practice — Area Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Area first, then cost
Trapezium: $(a + b) \div 2 \times h$

Question 5

(4 marks)

A garden is a trapezium. Its parallel sides are 12 m and 18 m, and they are 10 m apart. Turf costs £4.50 per m².

Work out the cost of turf for the garden.

..... £

Question 6

(3 marks)

A parallelogram has base 9 cm and height 4 cm. A triangle with base 12 cm has the same area.

Work out the height of the triangle.

..... cm

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$A = 9 \times 5$ M1

$= 45\ \text{cm}^2$ A1

Tip: base × perpendicular height.

Question 2

$A = \frac{1}{2} \times 14 \times 5$ M1

$= 35\ \text{cm}^2$ A1

Tip: 70 (no half) scores M0.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$A = \frac{1}{2} \times (6 + 10) \times 5$ **M1**

$= 40\ \text{cm}^2$ **A1**

Tip: 6 + 10 = 16, half of 16 is 8.

Question 4

$\frac{1}{2} \times 8 \times h = 36$ **M1**

$h = 9\ \text{cm}$ **A1**

Tip: $4h = 36$.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$A = \frac{1}{2} \times (12 + 18) \times 10$ — M1

$= 150\ \text{m}^2$ — A1

cost $= 150 \times 4.50$ — M1

$= £675$ — A1

Tip: find the area before the cost.

Question 6

parallelogram $= 9 \times 4 = 36\ \text{cm}^2$ — B1

$\frac{1}{2} \times 12 \times h = 36$ — M1

$h = 6$ cm — A1

Tip: $6h = 36$.